

Question	Marking Guidance	Mark	Comments
03.1	Correct answer of 12 = 2 marks OR Correct answer of 11.95 = 2 marks;; Incorrect answer but shows 1195 (ignoring orders of magnitude) = 1 mark OR Incorrect answer but shows 1.05 (Tg) or 13 (Tg) = 1 mark;	2 (2 x AO2)	Accept any numerical representation of the correct answer for two marks. Do not allow 1.05 or 13 to the incorrect order of magnitude e.g. $\times 10^1$.

Question	Marking Guidance	Mark	Comments
03.2	1. No significant difference (in production between untreated and treated) with no water shortage OR No significant difference (in production) between R and S; 2. Significantly greater/different (production in treated than untreated) with water shortage OR Significantly greater/different (production) in Q than P; 3. Significant decrease/difference (in production) of treated with water shortage OR Significant increase/difference (in production) of treated with no water shortage OR Significantly greater/different (production) in S than Q; 4. Overlap in SDs indicate no significant difference OR No overlap in SDs indicate a significant difference; 5. Only one (mycorrhizae) species studied; 6. Only shows (results for) tomato(es); 7. Large sample size increases validity/reliability; 8. (Investigation) done in sterile soil; 9. (Investigation) done in green/glasshouse OR Not done in field;	5 max (5 x AO3)	Max 3 marks from points 5 to 9 1, 2, 3 and 4. Accept descriptions of production e.g., more tomatoes and descriptions of treated and untreated seedlings / plants / seeds e.g., with / without mycorrhizae / <i>Glomus</i> / fungus. 1, 2, 3 and 4. Reject 'results are significant' or 'results are not significant' once, but only where there is no indication that these results are 'different', 'greater', 'reduced' etc. 1, 2, 3 and 4. Accept 'not due to chance' for significant and converse for not significant. 4. Accept '(error) bars' for SDs. 5. Accept 'only <i>Glomus</i> studied'. 6. Accept only shows 'one crop' or one 'species or type of plant'. 9. Do not credit idea of variation between green/glasshouses.

Question	Marking Guidance	Mark	Comments
03.3	1. Kill/remove/no (other) mycorrhizae/fungi; 2. Kill/remove/no nitrogen-fixing /nitrifying/denitrifying/saprobiotic bacteria; 3. Kill/remove/no pathogens; 4. Kill/remove/no pests; 5. Kill/remove/no competitors; 6. Kill/remove/no seeds/spores; 7. Is a (controlled) variable;	2 max (2 x AO2)	1 to 6. Accept in context of organisms being present if soil not sterilised. 2. Accept saprophytes and saprobionts. 3. Accept removes disease-causing bacteria/viruses /microorganisms. 3. Removes 'harmful bacteria' is not enough. 4. Accept named pests. 6. Accept weeds or (other) plants.

Question	Marking Guidance	Mark	Comments
03.4	1. So mycorrhizae/water is the only variable OR (So fertiliser) is not a variable OR So fertiliser is a controlled variable; 2. Affects (crop) production/growth/mass; 3. (Concentration of fertiliser) would affect water potential of soil; 4. Does not affect pH (of soil); 5. To provide (usual) soil/farming conditions;	2 max (2 x AO2)	Ignore leaching and eutrophication. 1. 'For comparison' on its own is not enough. 1. Accept same amount of fertiliser. 2. Accept benefits or limiting factor of named nutrient/ion.
